

Comparative Assessment: MRBENCH vs. MATHDIAL

Deployment context: Indian Metropolitan Grade 1–8 Mathematics Teachers (Delhi & Mumbai)

Deployment Context

Use case: Deployment scenario: A GPT-4 model is implemented in a one-on-one tutoring session, where its responses are augmented with a human tutor’s input. The tutor evaluates the pedagogical quality of the combined response, particularly in terms of how well it provides guidance. The goal is to assess whether the augmented GPT-4 responses are acceptable to the tutor.

Domain: Educational tutoring

Setting: Mobile application / enterprise software

Target population: High-end professional teacher teaching grade 1-8 in major metropolitan cities in India, such as Delhi and Mumbai, who is fluent in English.

Overall Risk Ratings

Benchmark	Risk	Avg. Score
MRBENCH	HIGH	3.2 / 5
MATHDIAL	HIGH	3.0 / 5

Dimension Scores Comparison

Dimension	MRBENCH	Rating	MATHDIAL	Rating	Δ
Input Ontology (IO)	4/5	Minor gaps	4/5	Minor gaps	0
Input Content (IC)	2/5	Significant gaps	1/5	Serious concern	+1
Input Form (IF)	4/5	Minor gaps	5/5	Strong alignment	-1
Output Ontology (OO)	3/5	Moderate gaps	2/5	Significant gaps	+1
Output Content (OC)	2/5	Significant gaps	1/5	Serious concern	+1
Output Form (OF)	4/5	Minor gaps	5/5	Strong alignment	-1
Average	3.2		3.0		+0.2

Δ = MRBENCH score – MATHDIAL score. Positive = regional benchmark scores higher.

Per-Dimension Justification Comparison

Brief excerpts from each benchmark's assessment justification. Full justifications are available in the individual dimension PDFs.

Input Ontology (IO)

Whether the benchmark's test case categories cover the query types expected in deployment.

MRBENCH — 4/5 (Minor gaps) [high confidence]	MATHDIAL — 4/5 (Minor gaps) [high confidence]
The user explicitly accepted curriculum-agnostic math errors as sufficient and confirmed the eight pedagogical dimensions are necessary and sufficient (elicitation Q1, Q3), reducing concern at the taxonomy level. The benchmark covers two difficulty levels (elementary and middle-school) [Q143, Q144], aligning with Grade 1–8 scope.	The user explicitly confirmed that curriculum-agnostic math errors and the benchmark's eight pedagogical categories are necessary and sufficient for the deployment task [A1, A3], substantially reducing ontological misalignment risk.

Input Content (IC)

Whether individual datapoint content is culturally and contextually appropriate for the target region.

MRBENCH — 2/5 (Significant gaps) [high confidence]	MATHDIAL — 1/5 (Serious concern) [high confidence]
This is a HIGH-priority dimension per elicitation. The benchmark content is drawn from Bridge [Q44–Q47] and MathDial [Q48, Q49], both Western-academic English-language datasets.	Input content is the highest-priority concern (user-flagged HIGH) and the dataset analysis confirms severe construct-irrelevant variance. The student simulation is calibrated to Western tutoring conventions [Q32, Q34] with documented limitations in representing human learning [Q106].

Input Form (IF)

Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

MRBENCH — 4/5 (Minor gaps) [high confidence]	MATHDIAL — 5/5 (Strong alignment) [high confidence]
The benchmark is text-only English [Q133, Q134, Q147], matching the deployment's text-based English mobile/enterprise application interface. The target population is fluent in English professionally.	MathDial is English text-only [Q1, Q32], and the deployment context is confirmed English text-only on mobile/enterprise platforms with no code-switching, script variation, or vernacular content. There is no modality, script, or language-encoding mismatch.

Output Ontology (OO)

Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

MRBENCH — 3/5 (Moderate gaps) [high confidence]	MATHDIAL — 2/5 (Significant gaps) [high confidence]
MODERATE priority per elicitation. The eight dimensions and three-tier labeling [Q58, Q59] are accepted by the user as necessary and sufficient (elicitation Q3). However, two structural concerns affect deployment validity.	User-flagged MODERATE risk. The user accepted the eight categories as necessary and sufficient [A3], reducing one source of ontological gap. However, the benchmark's output taxonomy operationalizes Equitable Tutoring [Q97, Q98, Q166] with an explicitly anti-telling, constructivist orientation, treating Telling as a last resort [Q155] and using Tell ...

Output Content (OC)

Whether ground-truth labels align with the judgments of regional stakeholders.

MRBENCH — 2/5 (Significant gaps) [high confidence]	MATHDIAL — 1/5 (Serious concern) [high confidence]
HIGH priority per elicitation. The annotation pool is four MBZUAI CS post-graduates [Q31] explicitly without required teaching experience [Q32], with no documented South Asian educational expertise.	User-flagged HIGH risk and the highest-priority validity concern. Ground-truth quality judgments were produced by 91 professional teachers recruited via Prolific [Q23], predominantly UK nationals, followed by USA, Canada, Australia, India, Germany [Q27] — India is the fifth of six named countries with no disaggregated count reported.

Output Form (OF)

Whether the expected output modality matches regional deployment needs and accessibility.

MRBENCH — 4/5 (Minor gaps) [high confidence]	MATHDIAL — 5/5 (Strong alignment) [high confidence]
LOWER priority per elicitation. Output form (categorical labels [Q58, Q59], DAMR percentages, and Pearson correlations [Q64, Q66]) is appropriate for the deployment scenario where teachers render acceptability judgments on AI+human-augmented responses.	Output form is well-aligned. Both benchmark and deployment evaluate text-based open-ended tutoring dialogue outputs. Automatic metrics (sBLEU, BERTScore, KF1, Uptake) [Q89, Q90, Q93] and interactive metrics (Success@k, Telling@k) [Q95, Q96] operate on text. Human evaluation uses 3-point Likert and binary scales [Q167] in standard formats.

Overall Summaries

MRBENCH (risk: HIGH)

MRBench is a well-constructed, narrowly-scoped benchmark for evaluating AI tutor responses to mathematics mistakes, with strengths in taxonomy validation, annotator training, and metric design. For the Delhi/Mumbai Grade 1–8 mathematics teacher deployment, validity is strongly preserved on dimensions involving curricular ontology (IO), input form (IF), and output form (OF), all of which align with the user's accepted text-only English, curriculum-agnostic deployment.

MATHDIAL (risk: HIGH)

MathDial presents a structurally sound but culturally narrow evaluation framework for AI tutoring quality. For the Indian Metropolitan Professional Teachers (Grades 1–8) deployment, three dimensions are well-aligned (Input Form, Output Form, and — given user confirmation that the eight categories suffice — Input Ontology), while three dimensions raise serious concerns.